# AUDITING ALGORITHMIC FAIRNESS AND GOVERNANCE IN CREDIT SCORING

*AI Generated Image

# Agenda


01 Introduction + Team (30s)

02 Data Quality Findings (90s)

03 Bias Analysis results (90s)

04 Governance recommendations

05 Conclusion

# 01 | Our Team

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# 02 DATA QUALITY

# 02 | Significant Data Quality Issues Were Identified Across Four Dimensions

## COMPLETENESS

*Missing values*

- 90 % loan_purpose values
- 87.6 % processing_timestamp
- 41.6% decision.approved_amount

## CONSISTENCY

*Inconsistent data formats*

- financials.annual_income stored as object instead of numeric
- Gender encoded inconsistently: Male / Female / M / F

## VALIDITY

*Invalid data values*

- credit_history_months min = -10
- savings_balance min = -5000
- 11 malformed email addresses detected

## UNIQUENESS

*Duplicate identifiers*

- 2 SSNs duplicated across 8 records

# 02 | A Rule-Based Cleaning Pipeline Standardized and Corrected the Dataset

### COMPLETENESS FIX

→ *Missing values standardized*

- Hidden missing tokens normalized (na, -, etc.)
- Missing loan purpose filled with “Unknown”

### CONSISTENCY FIX

→ *Data formats standardized*

- annual_income converted to numeric
- Gender standardized to Male / Female

### VALIDITY FIX

→ *Invalid values corrected*

- Negative financial values and incorrect formatted mail addresses are set to NaN

### UNIQUENESS FIX

→ *Identity risks flagged*

- Duplicate SSNs identified and flagged
- Records not automatically deleted

**Result: 500 records retained, 35 variables, Dataset ready for bias analysis**

# 03 BIAS ANALYSIS

# 03 | Female Applicants Experience Significantly Lower Approval Rates

| | Male | Female |
|---|---|---|
| Total Amount | 251 | 247 |
| Approved Count | 163 | 127 |
| Approval Rate | 65.9 % | 50.6% |
| Rejected Count | 124 | 84 |

$$\frac{50.6\%}{65.9\%} = 76.7\ \% \ \leq 80\%$$

Approval rate for female applicant falls below the Four-Fifths threshold, with a **gap of 15.4 p.p.,** and therefore indicates potential disparate impact.

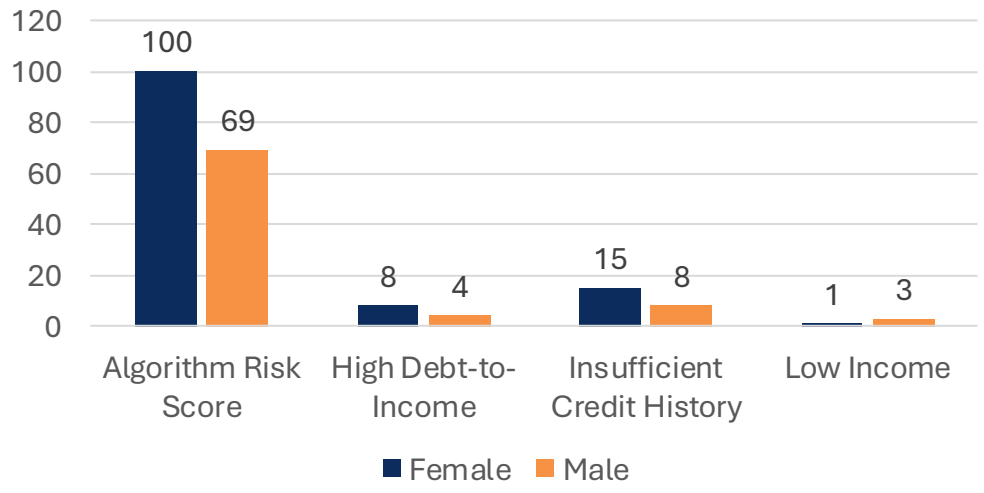
## Statistical Significance Testing

| | | |
|---|---|---|
| *Chi-Squared test of independence* | Chi² statistic: | 11.5053 |
| | P value: | **0.000694** |
| *Cramér's V* | Effect size | 0.1520 |

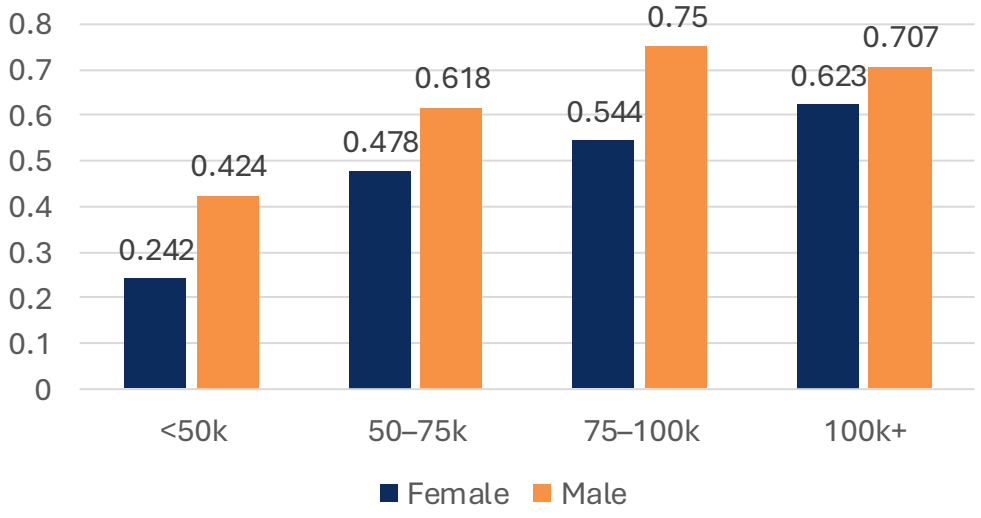
The chi-square test confirms a **statistically significant** relationship between gender and approval decisions ($p < 0.001$). The effect size (Cramér's V = 0.15) indicates a **small-to-moderate association**.

# 03 | Approval Disparity Persists Across Income Levels and Is Driven by Algorithmic Risk Scores

## Main Rejection Reasons




## Approval Rates by Income and Gender




**ACTION RECOMMENDATIONS**

Audit Risk Scoring Algorithm → Apply Fairness Constraints → Remove Gender Proxies → Implement Fairness Monitoring

# 04 Governance Recommendations

# 04 | Sensitive Personal Data Creates Re-Identification Risks

## Identified PII

### Direct Identifiers (DI)

- Full name
- Email
- SSN
- IP address
- Date of birth

### Quasi Identifiers (QI)

- Gender
- ZIP code
- Annual income

## Re-Identification Risk (k-anonymity)

| Metric | Value |
|---|---|
| Total records | 500 |
| Unique quasi-identifier combinations | 493 |
| Re-identification ratio | **98.6%** |
| Records with **k = 1** | **97.4%** |
| Records with **k ≥ 3** | **0%** |

97% of records have k = 1, meaning they can be uniquely identified. This represents a significant re-identification risk.

# 04 | Privacy Protection Implemented Through Pseudonymization and Data Generalization

## Privacy Techniques Applied

### Pseudonymization

- Email → SHA-256 hash
- SSN → SHA-256 hash
- Names removed
- IP addresses removed

### Generalization

- ZIP code → first 3 digits
- Date of birth → age bands

## k-Anonymity Improvement

| | Before | After |
|---|---|---|
| Unique combinations | 493 | **135** |
| Records with k = 1 | **97.4%** | **25.4%** |
| Records with k ≥ 3 | 0% | 0% |

After applying pseudonymization and generalization, the number of unique quasi-identifier combinations drops from 493 to 135. The share of uniquely identifiable records decreases from 97% to 25%, significantly reducing re-identification risk while preserving analytical value.

# 05 Conclusion

# 05 | Effective Credit AI Requires Integrated Data, Fairness, and Privacy Governance

## What Our Audit Revealed

- Data quality weaknesses can distort model inputs
- Approval decisions show statistically significant gender disparities
- Raw applicant data creates substantial re-identification risk

## Governance Implication

AI-based credit decisions must be governed across the **entire lifecycle**:

Data Collection
↓
Model Development
↓
Validation
↓
Deployment
↓
Monitoring

Responsible credit scoring requires continuous governance to ensure accurate data, fair decisions, and protected personal information.